

Lab steps to git clone vsdstdcelldesign

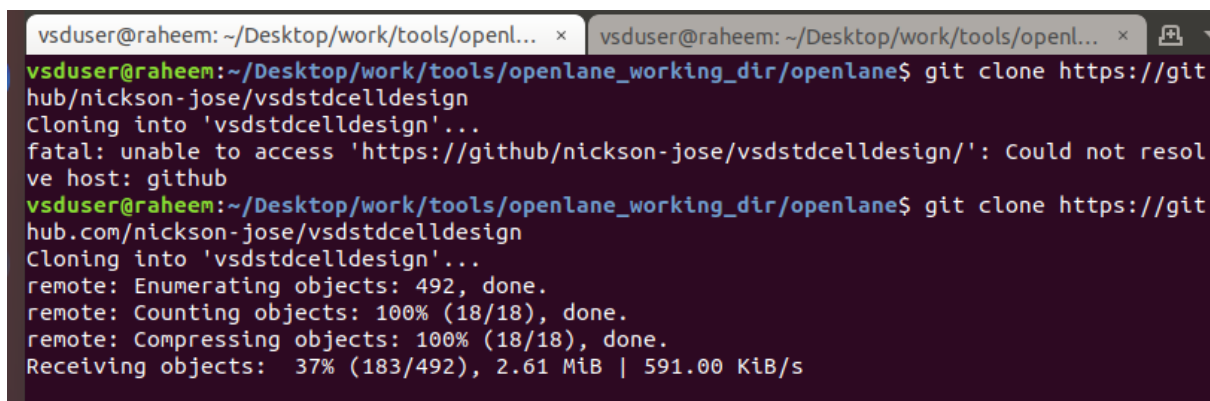
This lab guides you through using a custom inverter cell within the OpenLane design workflow. The aim is to clone a dedicated GitHub repository containing the necessary inverter files and SKY130 model libraries, configure your environment, set up the technology file, and visualize the inverter layout with Magic. Each step and command is documented to give a reproducible workflow for open-source hardware design.

Lab Steps

1. Clone the Custom Repository

Begin by cloning the provided GitHub repository, which contains the custom inverter designs and necessary model libraries.

git clone <https://github.com/nickson-jose/vsdstdcelldesign>

A terminal window showing the process of cloning a GitHub repository. The user is in a directory named 'openlane' and runs the command 'git clone https://github.com/nickson-jose/vsdstdcelldesign'. The terminal output shows the cloning progress, including enumerating objects, counting objects, and compressing objects. The final output shows the repository is cloned successfully.

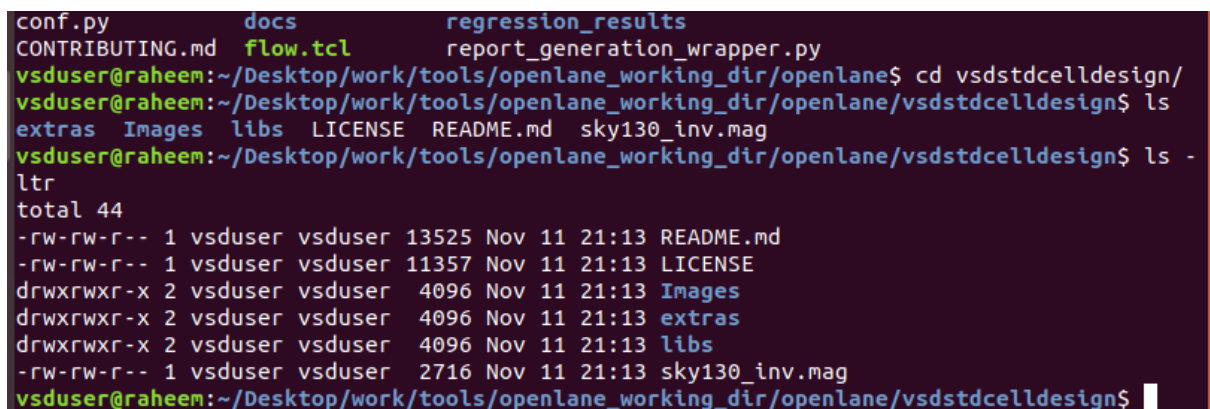
```
vsduser@raheem: ~/Desktop/work/tools/openlane... x vsduser@raheem: ~/Desktop/work/tools/openlane... x
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane$ git clone https://github.com/nickson-jose/vsdstdcelldesign
Cloning into 'vsdstdcelldesign'...
fatal: unable to access 'https://github.com/nickson-jose/vsdstdcelldesign/': Could not resolve host: github
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane$ git clone https://github.com/nickson-jose/vsdstdcelldesign
Cloning into 'vsdstdcelldesign'...
remote: Enumerating objects: 492, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (18/18), done.
Receiving objects: 37% (183/492), 2.61 MiB | 591.00 KiB/s
```

Verify Repository Contents

Navigate into the cloned directory and inspect the files to ensure all resources, including Mac files and model libraries, are present.

cd vsdstdcelldesign

ls

A terminal window showing the contents of the cloned repository. The user is in the 'vsdstdcelldesign' directory and runs the command 'ls'. The terminal output shows a list of files and directories, including 'conf.py', 'docs', 'regression_results', 'CONTRIBUTING.md', 'flow.tcl', 'report_generation_wrapper.py', 'extras', 'Images', 'libs', 'LICENSE', 'README.md', and 'sky130_inv.mag'. The user then runs 'ls -ltr' to show the files in long format, sorted by time.

```
conf.py      docs        regression_results
CONTRIBUTING.md  flow.tcl    report_generation_wrapper.py
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane$ cd vsdstdcelldesign/
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$ ls
extras Images libs LICENSE README.md sky130_inv.mag
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$ ls -ltr
total 44
-rw-rw-r-- 1 vsduser vsduser 13525 Nov 11 21:13 README.md
-rw-rw-r-- 1 vsduser vsduser 11357 Nov 11 21:13 LICENSE
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 Images
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 extras
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 libs
-rw-rw-r-- 1 vsduser vsduser 2716 Nov 11 21:13 sky130_inv.mag
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$
```

Locate the Magic Tech File

Before opening layout files, ensure you have the appropriate Magic technology file (.tech) in your working directory. The file is typically inside another project or PDK folder.

Navigate to the directory containing the tech file (replace paths as appropriate):

```
cd
/home/vsduser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech/magic
ls *.tech
```

```
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks$ cd sky130A/
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A$ ls
libs.ref  libs.tech  SOURCES
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A$ cd libs.tech/
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech$ magic
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech$ ls
irsim  klayout  magic  netgen  ngspice  openlane  qflow  xcircuit  xschem
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech$
```

Copy the Magic Tech File

Copy the required tech file from its location into your current project directory.

```
cp
/home/vsduser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech/magic/sky130A.tech
/home/vsduser/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign
```

```
generate_fclt.py sky130A-dbs.tech sky130A.tech
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech/magic$ c
p sky130A.tech /home/vsduser/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcell
design/
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech/magic$
```

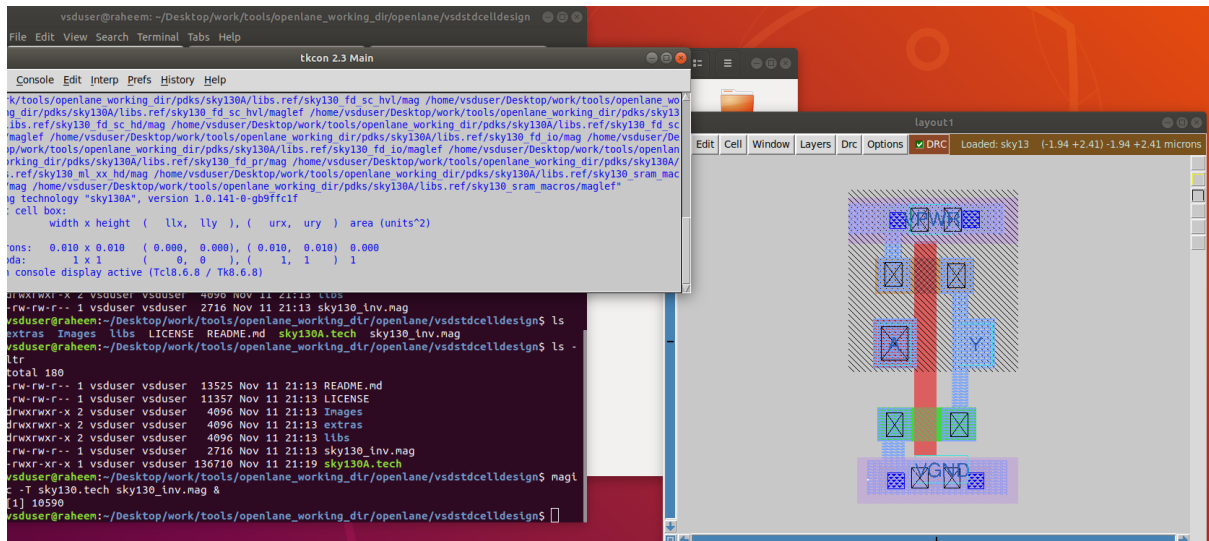
```
extras  images  tcl  LICENSE  README.md  sky130A.tech  sky130A_inv.mag
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$ ls
ltr
total 180
-rw-rw-r-- 1 vsduser vsduser 13525 Nov 11 21:13 README.md
-rw-rw-r-- 1 vsduser vsduser 11357 Nov 11 21:13 LICENSE
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 Images
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 extras
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 libs
-rw-rw-r-- 1 vsduser vsduser 2716 Nov 11 21:13 sky130_inv.mag
-rwxr-xr-x 1 vsduser vsduser 136710 Nov 11 21:19 sky130A.tech
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$
```

Open the Inverter Layout in Magic

Launch Magic using the local tech file and open the inverter schematic or layout file (.mag). The & at the end of the command frees the terminal for additional commands.

```
magic -T sky130A.tech sky130_inv.mag &
```

- Replace sky130_inv.mag with your inverter's .mag file if it has a different name.



Explore the Layout in Magic

Once Magic opens:

- Move your cursor to the layout window.
- Use the s key to select.
- Use the v key to centralize or zoom into the selected area.

